<u>ADDITIONAL WORK ADDRESSING ICML REVIEWERS' QUERIES</u>

Experimental Results

We have conducted 5-fold cross validation for each of the 5 datasets (Breast Cancer, OSHA, Red Wine, Pulsar and Diabetes), each fold repeated 10 times i.e. 50 runs of each experiment have been conducted. Our proposed model (CSVM) has been compared with 4 SVM kernels (RBF, Sigmoid, Polynomial), 2 whitening algorithms (ZCA whitening + SVM, PCA whitening + SVM) and 2 models incorporating class covariances suggested by the reviewers (Minimum class variance support vector machines (MCVSVM) and minimax probability machines (MPM). Also CSVM has been compared with metric model Neighborhood component Analysis (NCA)

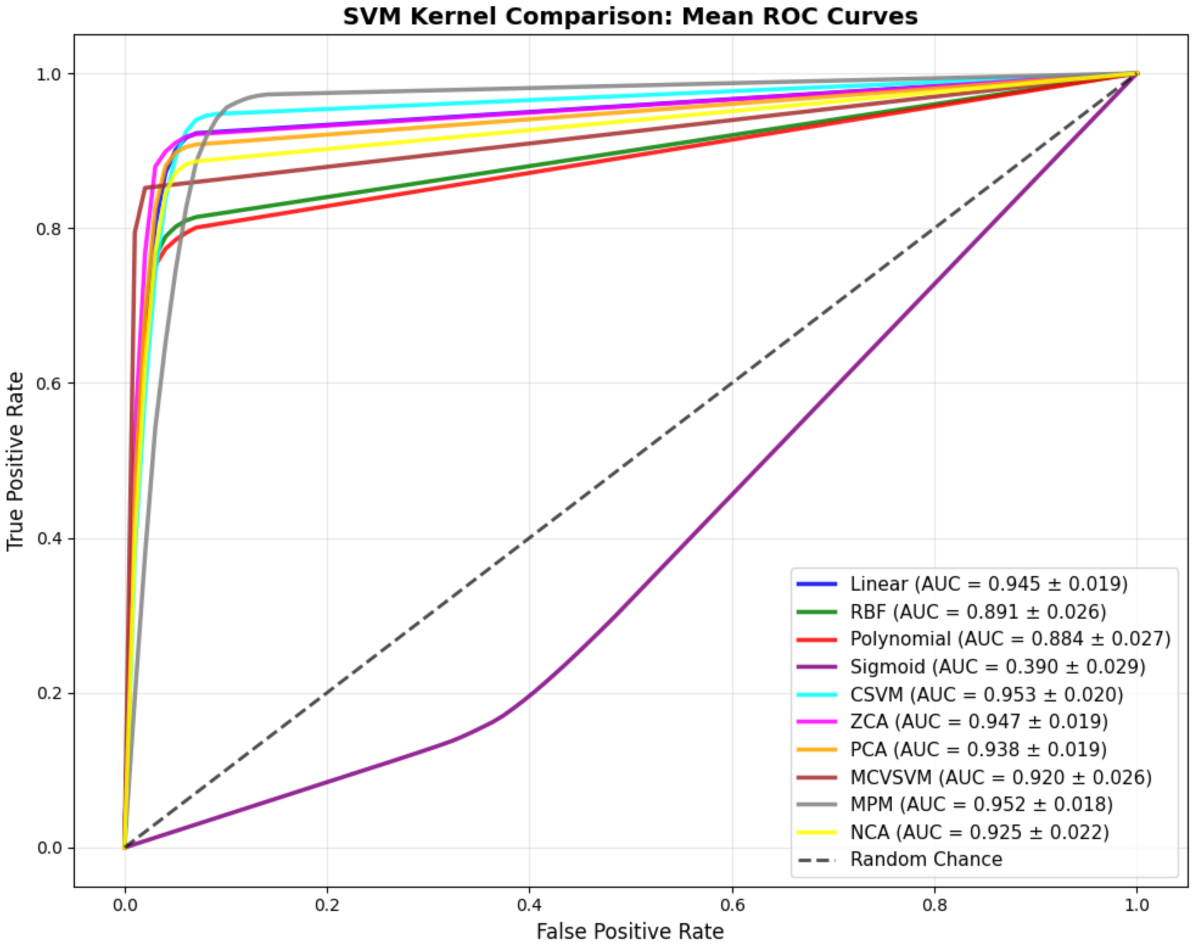
The results are as follows:

a) Breast Cancer Dataset

a. Classification Table of CSVM vs Baseline Models

	Accuracy	Precision	Recall	F1_Score
CSVM	**95.64%**	95.39%	**95.39%**	**95.34%**
Linear SVM	95.29%	95.38%	94.60%	94.92%
RBF SVM	91.49%	92.90%	89.24%	90.52%
Sigmoid SVM	44.90%	37.57%	39.01%	38.02%
Polynomial SVM	90.91%	92.44%	88.48%	89.83%
ZCA-SVM	95.61%	**95.84%**	94.81%	95.24%
PCA-SVM	94.85%	95.11%	93.93%	94.42%
MCVSVM	94.18%	95.59%	92.31%	93.54%
MPM	94.78%	94.08%	95.23%	94.51%
NCA	93.74%	94.05%	92.60%	93.19%

b. ROC Curve



c. Statistical significance tests

i. Statistical significance of Accuracy of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.9564	0.9529	0.0035	3.66E-03	Yes
CSVM vs RBF	0.9564	0.9149	0.0415	3.03E-15	Yes
CSVM vs Sigmoid	0.9564	0.449	0.5074	4.25E-57	Yes
CSVM vs Poly	0.9564	0.9091	0.0473	1.51E-17	Yes
CSVM vs ZCA	0.9564	0.9561	0.0003	9.16E-01	No
CSVM vs PCA	0.9564	0.9485	0.0079	1.50E-03	Yes
CSVM vs MCVM	0.9564	0.9418	0.0146	4.75E-05	Yes
CSVM vs MPM	0.9564	0.9478	0.0086	2.70E-02	Yes
CSVM vs NCA	0.9564	0.9374	0.019	3.75E-07	Yes

ii. Statistical significance of Precision of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.9539	0.9538	0.0001	9.55E-01	No
CSVM vs RBF	0.9539	0.929	0.0249	7.60E-09	Yes
CSVM vs Sigmoid	0.9539	0.3757	0.5782	7.95E-58	Yes
CSVM vs Poly	0.9539	0.9244	0.0295	1.17E-10	Yes
CSVM vs ZCA	0.9539	0.9584	-0.0045	1.91E-01	No
CSVM vs PCA	0.9539	0.9511	0.0028	2.36E-01	No
CSVM vs MCVM	0.9539	0.9559	-0.002	5.31E-01	No
CSVM vs MPM	0.9539	0.9408	0.0132	2.37E-03	Yes
CSVM vs NCA	0.9539	0.9405	0.0134	5.04E-04	Yes

iii. Statistical significance of Recall of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.9539	0.946	0.008	2.52E-06	Yes
CSVM vs RBF	0.9539	0.8923	0.0616	9.56E-19	Yes
CSVM vs Sigmoid	0.9539	0.3901	0.5639	7.83E-59	Yes
CSVM vs Poly	0.9539	0.8848	0.0691	3.32E-21	Yes
CSVM vs ZCA	0.9539	0.9481	0.0058	1.30E-01	No
CSVM vs PCA	0.9539	0.9393	0.0146	3.45E-06	Yes
CSVM vs MCVM	0.9539	0.9231	0.0309	3.36E-10	Yes
CSVM vs MPM	0.9539	0.9523	0.0016	6.68E-01	No
CSVM vs NCA	0.9539	0.926	0.0279	2.99E-10	Yes

iv. Statistical significance of F1 Scores of CSVM vs baseline models

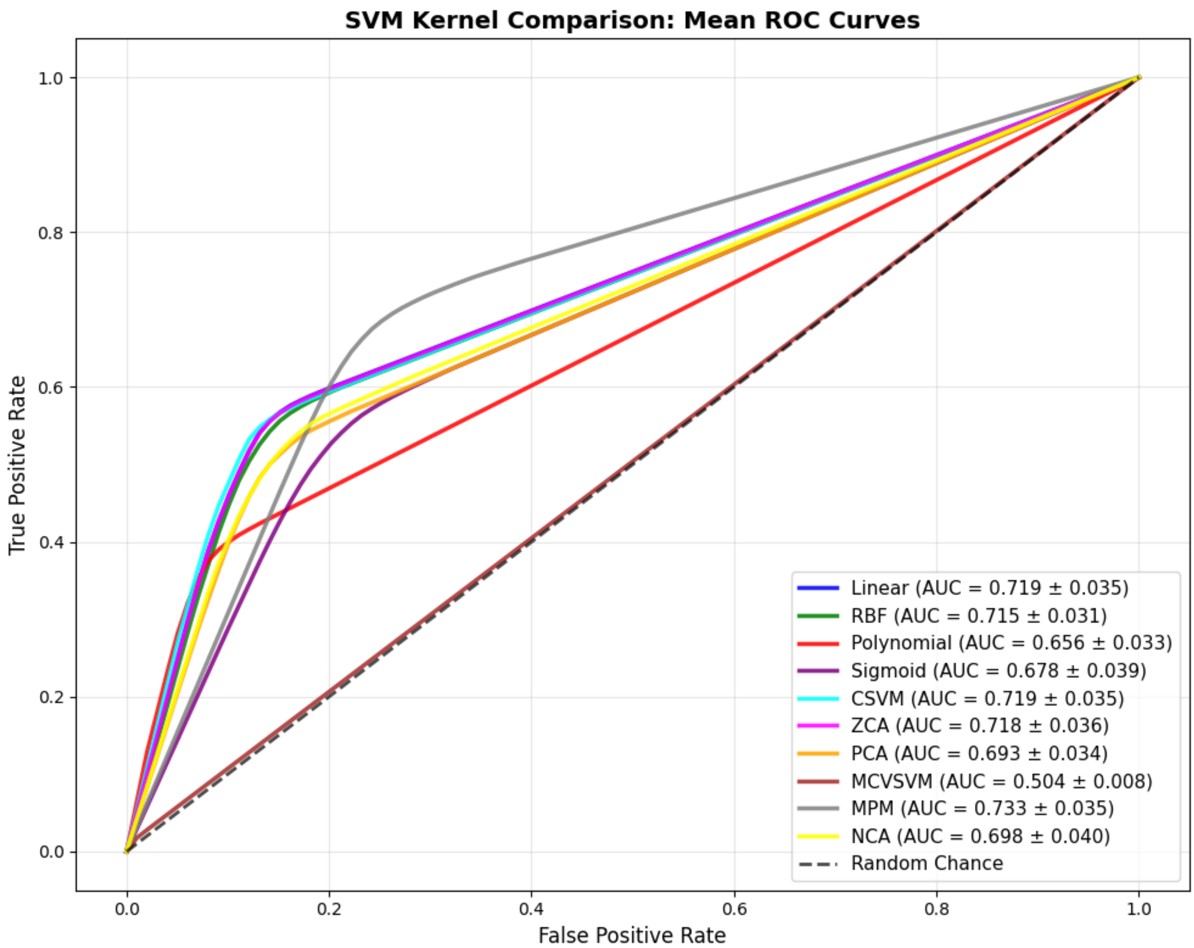
Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.9534	0.9492	0.0042	1.53E-03	Yes
CSVM vs RBF	0.9534	0.9052	0.0482	8.09E-16	Yes
CSVM vs Sigmoid	0.9534	0.3802	0.5732	7.43E-59	Yes
CSVM vs Poly	0.9534	0.8983	0.0551	4.77E-18	Yes
CSVM vs ZCA	0.9534	0.9524	0.001	7.85E-01	No
CSVM vs PCA	0.9534	0.9442	0.0092	7.08E-04	Yes
CSVM vs MCVM	0.9534	0.9354	0.018	8.59E-06	Yes
CSVM vs MPM	0.9534	0.9451	0.0083	4.18E-02	Yes
CSVM vs NCA	0.9534	0.9319	0.0215	1.31E-07	Yes

b) Diabetes Dataset

a. Classification Table of CSVM vs Baseline Models

	Accuracy	Precision	Recall	F1_Score
CSVM	**77.00%**	**75.68%**	71.88%	**72.90%**
Linear SVM	76.68%	75.04%	71.87%	72.78%
RBF SVM	76.24%	74.50%	71.46%	72.32%
Sigmoid SVM	71.28%	68.43%	67.76%	67.95%
Polynomial SVM	73.84%	74.04%	65.63%	66.34%
ZCA-SVM	76.65%	75.01%	71.84%	72.74%
PCA-SVM	74.49%	72.41%	69.28%	70.04%
MCVSVM	65.33%	46.66%	50.40%	40.40%
MPM	74.20%	72.21%	**73.29%**	72.43%
NCA	74.82%	72.80%	69.85%	70.50%

b. ROC Curve



c. Statistical significance tests

i. Statistical significance of Accuracy of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.77	0.7668	0.0033	2.50E-03	Yes
CSVM vs RBF	0.77	0.7624	0.0077	1.97E-02	Yes
CSVM vs Sigmoid	0.77	0.7128	0.0573	5.54E-14	Yes
CSVM vs Poly	0.77	0.7384	0.0316	7.80E-12	Yes
CSVM vs ZCA	0.77	0.7665	0.0035	1.18E-03	Yes
CSVM vs PCA	0.77	0.7449	0.0251	9.89E-08	Yes
CSVM vs MCVM	0.77	0.6533	0.1168	2.19E-30	Yes
CSVM vs MPM	0.77	0.742	0.028	4.37E-09	Yes
CSVM vs NCA	0.77	0.7482	0.0219	3.84E-06	Yes

ii. Statistical significance of Precision of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7568	0.7504	0.0064	1.40E-05	Yes
CSVM vs RBF	0.7568	0.745	0.0118	3.53E-03	Yes
CSVM vs Sigmoid	0.7568	0.6843	0.0725	3.07E-15	Yes
CSVM vs Poly	0.7568	0.7404	0.0164	1.76E-03	Yes
CSVM vs ZCA	0.7568	0.7501	0.0067	6.83E-06	Yes
CSVM vs PCA	0.7568	0.7241	0.0326	3.11E-08	Yes
CSVM vs MCVM	0.7568	0.4666	0.2902	2.88E-12	Yes
CSVM vs MPM	0.7568	0.7221	0.0347	1.06E-10	Yes
CSVM vs NCA	0.7568	0.728	0.0288	3.77E-06	Yes

iii. Statistical significance of Recall of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7188	0.7187	0.0001	9.32E-01	No
CSVM vs RBF	0.7188	0.7146	0.0042	2.62E-01	No
CSVM vs Sigmoid	0.7188	0.6776	0.0412	1.56E-08	Yes
CSVM vs Poly	0.7188	0.6563	0.0625	4.32E-19	Yes
CSVM vs ZCA	0.7188	0.7184	0.0005	6.64E-01	No
CSVM vs PCA	0.7188	0.6928	0.0261	5.05E-06	Yes
CSVM vs MCVM	0.7188	0.504	0.2149	9.49E-39	Yes

CSVM vs MPM	0.7188	0.7329	-0.0141	2.00E-03	Yes
CSVM vs NCA	0.7188	0.6985	0.0203	3.01E-04	Yes

iv. Statistical significance of F1 Scores of CSVM vs baseline models

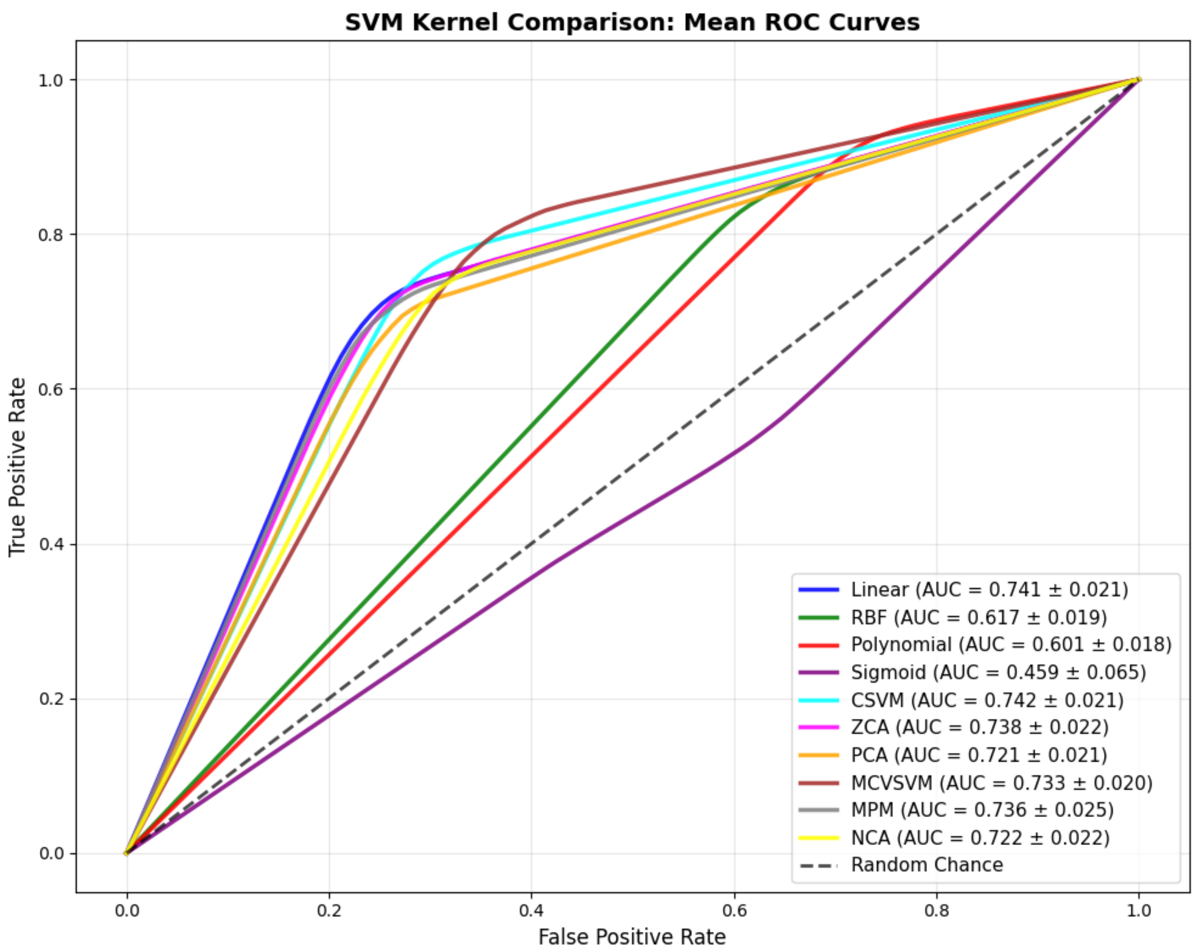
Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.729	0.7278	0.0012	3.28E-01	No
CSVM vs RBF	0.729	0.7232	0.0057	1.40E-01	No
CSVM vs Sigmoid	0.729	0.6795	0.0495	2.26E-10	Yes
CSVM vs Poly	0.729	0.6634	0.0656	4.05E-18	Yes
CSVM vs ZCA	0.729	0.7274	0.0015	1.82E-01	No
CSVM vs PCA	0.729	0.7004	0.0286	1.35E-06	Yes
CSVM vs MCVM	0.729	0.404	0.325	8.19E-44	Yes
CSVM vs MPM	0.729	0.7243	0.0047	3.00E-01	No
CSVM vs NCA	0.729	0.705	0.0239	9.04E-05	Yes

c) Red Wine Dataset

a. Classification Table of CSVM vs Baseline Models

	Accuracy	Precision	Recall	F1_Score
CSVM	**74.39%**	**74.34%**	**74.23%**	**74.23%**
Linear SVM	73.96%	74.08%	74.12%	73.92%
RBF SVM	63.43%	65.64%	61.74%	60.13%
Sigmoid SVM	46.18%	45.87%	45.90%	45.84%
Polynomial SVM	62.42%	68.28%	60.15%	56.44%
ZCA-SVM	73.68%	73.74%	73.77%	73.61%
PCA-SVM	71.94%	72.06%	72.11%	71.90%
MCVSVM	73.89%	74.29%	73.35%	73.40%
MPM	73.41%	73.55%	73.57%	73.37%
NCA	72.28%	72.21%	72.16%	72.14%

b. ROC Curve



c. Statistical significance tests

i. Statistical significance of Accuracy of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7439	0.7396	0.0043	7.36E-03	Yes
CSVM vs RBF	0.7439	0.6343	0.1096	2.28E-35	Yes
CSVM vs Sigmoid	0.7439	0.4618	0.2821	4.03E-32	Yes
CSVM vs Poly	0.7439	0.6242	0.1197	2.06E-35	Yes
CSVM vs ZCA	0.7439	0.7368	0.0071	2.58E-05	Yes
CSVM vs PCA	0.7439	0.7194	0.0245	8.09E-15	Yes
CSVM vs MCVM	0.7439	0.7389	0.005	2.54E-02	Yes
CSVM vs MPM	0.7439	0.7341	0.0098	1.60E-06	Yes
CSVM vs NCA	0.7439	0.7228	0.0211	1.59E-05	Yes

ii. Statistical significance of Precision of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7434	0.7408	0.0026	8.93E-02	No
CSVM vs RBF	0.7434	0.6564	0.087	1.33E-29	Yes
CSVM vs Sigmoid	0.7434	0.4587	0.2847	4.53E-32	Yes
CSVM vs Poly	0.7434	0.6828	0.0607	1.83E-17	Yes
CSVM vs ZCA	0.7434	0.7374	0.006	2.54E-04	Yes
CSVM vs PCA	0.7434	0.7206	0.0228	5.10E-14	Yes
CSVM vs MCVM	0.7434	0.7429	0.0005	7.89E-01	No
CSVM vs MPM	0.7434	0.7355	0.008	7.07E-05	Yes
CSVM vs NCA	0.7434	0.7221	0.0213	1.58E-05	Yes

iii. Statistical significance of Recall of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7423	0.7412	0.0011	4.76E-01	No
CSVM vs RBF	0.7423	0.6174	0.1249	1.43E-37	Yes
CSVM vs Sigmoid	0.7423	0.459	0.2833	4.69E-32	Yes
CSVM vs Poly	0.7423	0.6015	0.1408	1.68E-38	Yes
CSVM vs ZCA	0.7423	0.7377	0.0046	3.68E-03	Yes
CSVM vs PCA	0.7423	0.7211	0.0212	1.61E-12	Yes
CSVM vs MCVM	0.7423	0.7335	0.0088	5.16E-04	Yes

CSVM vs MPM	0.7423	0.7357	0.0065	7.41E-04	Yes
CSVM vs NCA	0.7423	0.7216	0.0207	2.09E-05	Yes

iv. Statistical significance of F1 Scores of CSVM vs baseline models

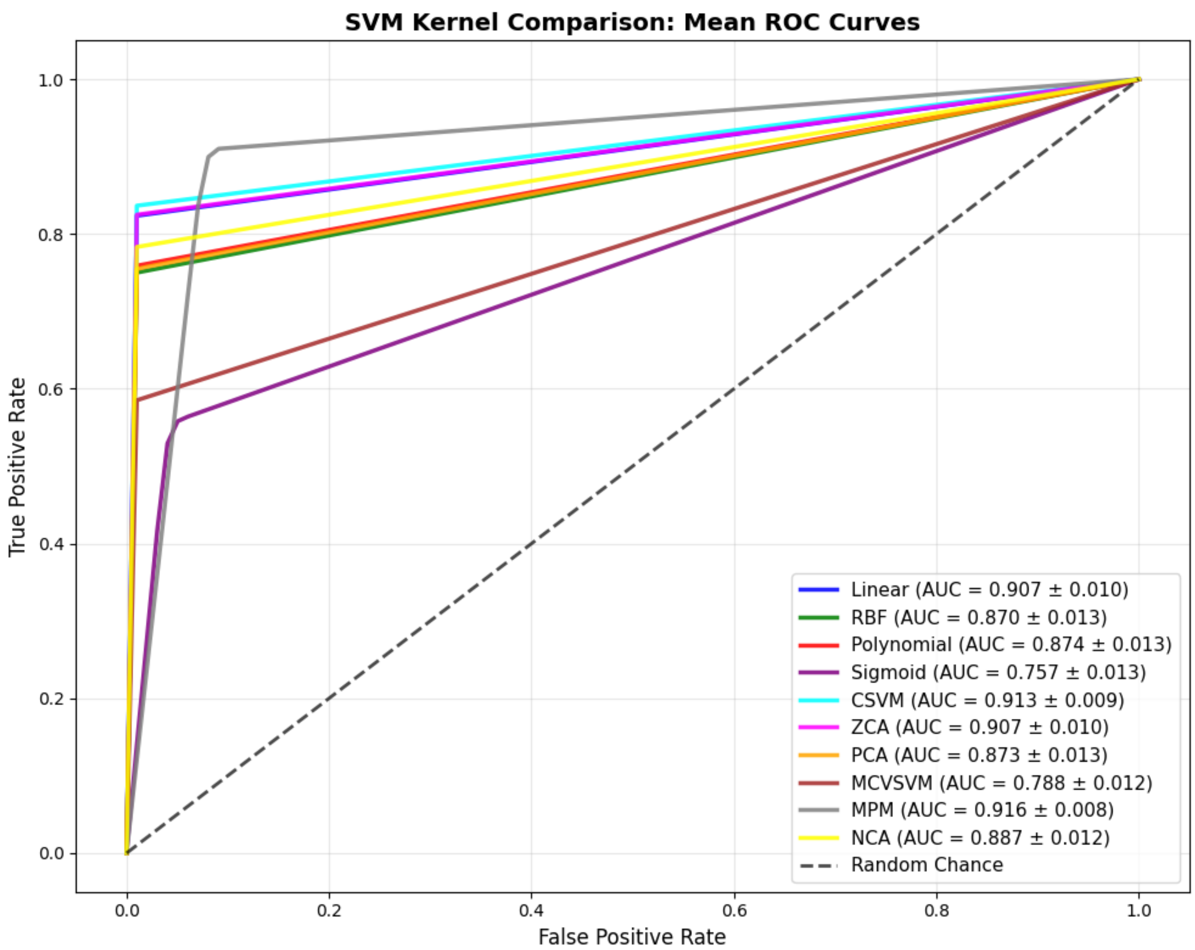
Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7423	0.7392	0.003	5.27E-02	No
CSVM vs RBF	0.7423	0.6013	0.141	2.57E-37	Yes
CSVM vs Sigmoid	0.7423	0.4584	0.2838	4.54E-32	Yes
CSVM vs Poly	0.7423	0.5644	0.1778	4.11E-40	Yes
CSVM vs ZCA	0.7423	0.7361	0.0061	2.22E-04	Yes
CSVM vs PCA	0.7423	0.719	0.0232	9.05E-14	Yes
CSVM vs MCVM	0.7423	0.734	0.0082	1.31E-03	Yes
CSVM vs MPM	0.7423	0.7337	0.0086	1.77E-05	Yes
CSVM vs NCA	0.7423	0.7214	0.0209	1.82E-05	Yes

d) Pulsar Dataset

a. Classification Table of CSVM vs Baseline Models

	Accuracy	Precision	Recall	F1_Score
CSVM	**97.98%**	96.04%	91.51%	**93.62%**
Linear SVM	97.91%	96.24%	90.88%	93.35%
RBF SVM	97.26%	95.73%	87.20%	90.90%
Sigmoid SVM	92.21%	76.71%	75.70%	76.16%
Polynomial SVM	97.20%	94.93%	87.59%	90.84%
ZCA-SVM	97.92%	96.22%	90.95%	93.38%
PCA-SVM	97.19%	95.03%	87.40%	90.76%
MCVSVM	96.09%	**97.31%**	79.02%	85.51%
MPM	92.19%	76.74%	**91.60%**	81.82%
NCA	97.49%	95.55%	88.83%	91.84%

b. ROC Curve



c. Statistical significance tests

i. Statistical significance of Accuracy of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.9798	0.9791	0.0007	3.69E-07	Yes
CSVM vs RBF	0.9798	0.9726	0.0073	5.71E-33	Yes
CSVM vs Sigmoid	0.9798	0.9221	0.0578	1.51E-52	Yes
CSVM vs Poly	0.9798	0.972	0.0078	1.64E-32	Yes
CSVM vs ZCA	0.9798	0.9792	0.0006	1.73E-07	Yes
CSVM vs PCA	0.9798	0.9719	0.008	5.00E-35	Yes
CSVM vs MCVM	0.9798	0.9609	0.0189	2.53E-46	Yes
CSVM vs MPM	0.9798	0.9219	0.0579	3.46E-51	Yes
CSVM vs NCA	0.9798	0.9749	0.005	7.79E-25	Yes

ii. Statistical significance of Precision of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.9604	0.9624	-0.002	1.19E-07	Yes
CSVM vs RBF	0.9604	0.9573	0.0031	2.58E-05	Yes
CSVM vs Sigmoid	0.9604	0.7671	0.1934	2.82E-56	Yes
CSVM vs Poly	0.9604	0.9493	0.0112	1.44E-18	Yes
CSVM vs ZCA	0.9604	0.9622	-0.0018	5.56E-08	Yes
CSVM vs PCA	0.9604	0.9503	0.0102	2.73E-20	Yes
CSVM vs MCVM	0.9604	0.9731	-0.0127	3.94E-20	Yes
CSVM vs MPM	0.9604	0.7674	0.193	1.29E-63	Yes
CSVM vs NCA	0.9604	0.9555	0.0049	1.05E-05	Yes

iii. Statistical significance of Recall of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.9151	0.9088	0.0063	4.60E-17	Yes
CSVM vs RBF	0.9151	0.872	0.0431	3.68E-36	Yes
CSVM vs Sigmoid	0.9151	0.757	0.1581	6.09E-54	Yes
CSVM vs Poly	0.9151	0.8759	0.0393	1.71E-33	Yes
CSVM vs ZCA	0.9151	0.9095	0.0056	5.84E-17	Yes
CSVM vs PCA	0.9151	0.874	0.0411	9.04E-37	Yes
CSVM vs MCVM	0.9151	0.7902	0.1249	4.40E-53	Yes

CSVM vs MPM	0.9151	0.916	-0.0009	4.95E-01	No
CSVM vs NCA	0.9151	0.8883	0.0268	3.67E-28	Yes

iv. Statistical significance of F1 Scores of CSVM vs baseline models

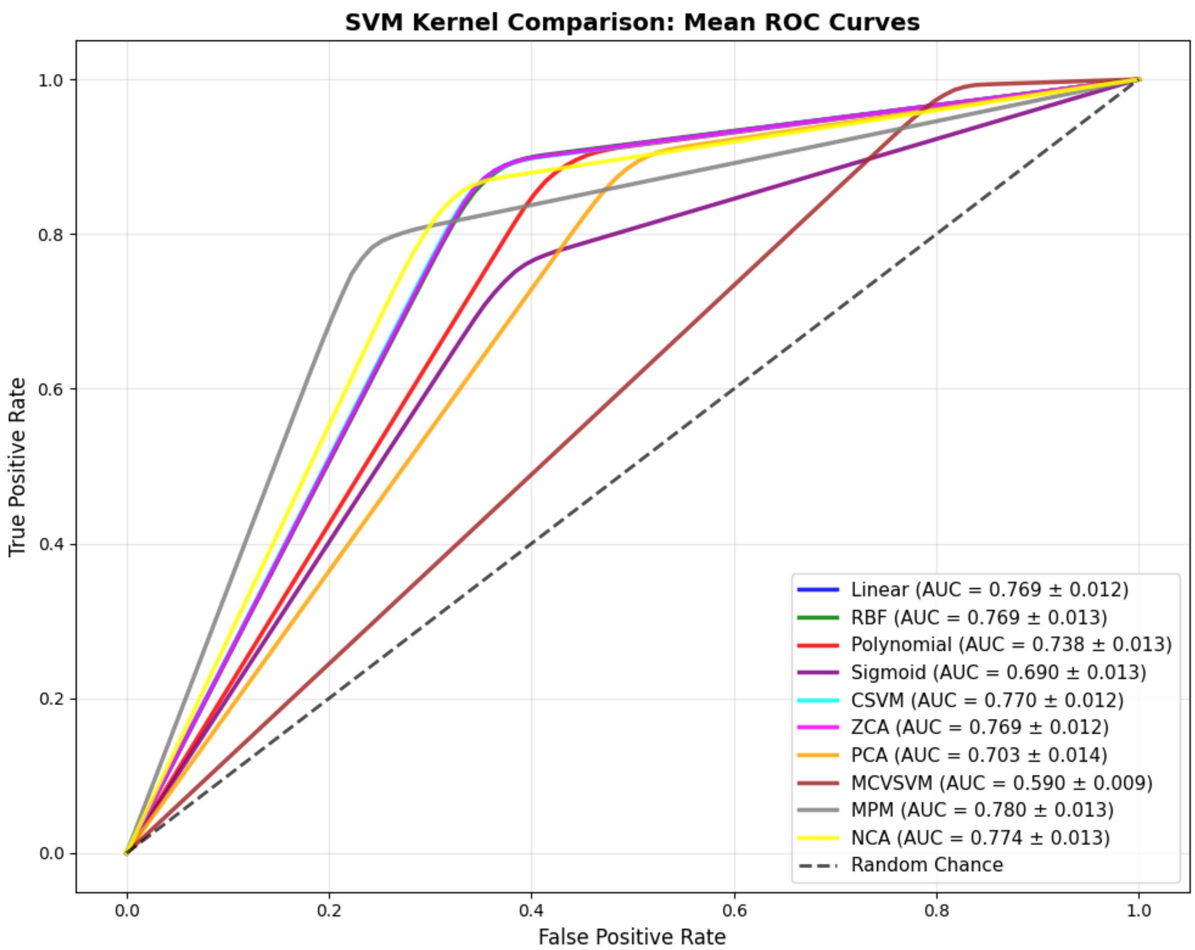
Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.9362	0.9335	0.0028	4.29E-09	Yes
CSVM vs RBF	0.9362	0.909	0.0272	3.11E-33	Yes
CSVM vs Sigmoid	0.9362	0.7616	0.1746	1.46E-56	Yes
CSVM vs Poly	0.9362	0.9084	0.0279	2.01E-32	Yes
CSVM vs ZCA	0.9362	0.9338	0.0025	1.76E-09	Yes
CSVM vs PCA	0.9362	0.9076	0.0286	6.07E-35	Yes
CSVM vs MCVM	0.9362	0.8551	0.0811	3.03E-47	Yes
CSVM vs MPM	0.9362	0.8182	0.118	1.23E-51	Yes
CSVM vs NCA	0.9362	0.9184	0.0178	6.78E-26	Yes

e) OSHA Dataset

a. Classification Table of CSVM vs Baseline Models

	Accuracy	Precision	Recall	F1_Score
CSVM	**79.66%**	79.48%	76.99%	77.76%
Linear SVM	79.64%	79.51%	76.93%	77.72%
RBF SVM	79.65%	79.56%	76.89%	77.69%
Sigmoid SVM	70.61%	69.09%	69.05%	69.05%
Polynomial SVM	77.43%	77.85%	73.78%	74.69%
ZCA-SVM	79.64%	79.50%	76.92%	77.71%
PCA-SVM	74.75%	75.40%	70.30%	71.06%
MCVSVM	67.97%	**79.71%**	59.02%	55.23%
MPM	78.24%	77.21%	**77.96%**	77.45%
NCA	79.34%	78.60%	77.38%	**77.83%**

b. ROC Curve



c. Statistical significance tests

i. Statistical significance of Accuracy of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7966	0.7964	0.0002	4.24E-01	No
CSVM vs RBF	0.7966	0.7965	0.0001	9.17E-01	No
CSVM vs Sigmoid	0.7966	0.7061	0.0904	3.10E-46	Yes
CSVM vs Poly	0.7966	0.7743	0.0223	1.29E-21	Yes
CSVM vs ZCA	0.7966	0.7964	0.0002	3.12E-01	No
CSVM vs PCA	0.7966	0.7475	0.0491	2.10E-35	Yes
CSVM vs MCVM	0.7966	0.6797	0.1169	3.52E-49	Yes
CSVM vs MPM	0.7966	0.7824	0.0142	1.17E-12	Yes
CSVM vs NCA	0.7966	0.7934	0.0032	1.76E-02	Yes

ii. Statistical significance of Precision of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7948	0.7951	-0.0002	2.89E-01	No
CSVM vs RBF	0.7948	0.7956	-0.0008	4.85E-01	No
CSVM vs Sigmoid	0.7948	0.6909	0.1039	1.41E-47	Yes
CSVM vs Poly	0.7948	0.7785	0.0164	3.92E-13	Yes
CSVM vs ZCA	0.7948	0.795	-0.0002	4.05E-01	No
CSVM vs PCA	0.7948	0.754	0.0408	1.07E-27	Yes
CSVM vs MCVM	0.7948	0.7971	-0.0023	2.84E-01	No
CSVM vs MPM	0.7948	0.7721	0.0228	1.83E-20	Yes
CSVM vs NCA	0.7948	0.786	0.0089	2.07E-07	Yes

iii. Statistical significance of Recall of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7699	0.7693	0.0006	2.76E-02	Yes
CSVM vs RBF	0.7699	0.7689	0.001	3.55E-01	No
CSVM vs Sigmoid	0.7699	0.6905	0.0794	8.19E-42	Yes
CSVM vs Poly	0.7699	0.7378	0.032	3.04E-27	Yes
CSVM vs ZCA	0.7699	0.7692	0.0006	1.42E-02	Yes
CSVM vs PCA	0.7699	0.703	0.0669	1.29E-38	Yes
CSVM vs MCVM	0.7699	0.5902	0.1797	1.18E-54	Yes

CSVM vs MPM	0.7699	0.7796	-0.0097	1.17E-07	Yes
CSVM vs NCA	0.7699	0.7738	-0.0039	6.13E-03	Yes

iv. Statistical significance of F1 Scores of CSVM vs baseline models

Comparison	CSVM Mean	Baseline Mean	Mean Difference	t-test p-value	Significant?
CSVM vs Linear	0.7776	0.7772	0.0004	8.79E-02	No
CSVM vs RBF	0.7776	0.7769	0.0007	5.43E-01	No
CSVM vs Sigmoid	0.7776	0.6905	0.0871	6.34E-44	Yes
CSVM vs Poly	0.7776	0.7469	0.0307	2.15E-25	Yes
CSVM vs ZCA	0.7776	0.7771	0.0005	5.27E-02	No
CSVM vs PCA	0.7776	0.7106	0.067	1.24E-37	Yes
CSVM vs MCVM	0.7776	0.5523	0.2253	5.24E-54	Yes
CSVM vs MPM	0.7776	0.7745	0.0031	6.17E-02	No
CSVM vs NCA	0.7776	0.7783	-0.0007	6.34E-01	No

Vector space transformation to Euclidean space for SVM

Definition 2.1. ***The classical Support Vector Machine assumes an identity covariance matrix. While the Karush-Kuhn-Tucker (KKT) conditions provide a unique global L2-norm based optimal solution, this geometric margin is statistically sub-optimal when data exhibits significant class-specific non-identical covariance structures ($\Sigma_1 \neq \Sigma_2 \neq I$). In such "Non-Euclidean" statistical manifolds, the KKT conditions optimize for a geometric separation that does not coincide with the Bayes optimal decision boundary.***

Proof: For a probability distribution of a population, distance between a point and the mean in statistical/input space given by the Mahalanobis distance written as:

$$(\mathrm{X}-\mu)^{\mathrm{T}}\Sigma^{-1}(\mathrm{X}-\mu) \tag{1}$$

Mahalanobis distance can be written as:

$$(\mathrm{X}-\mu)^{\mathrm{T}}\Sigma^{-1}(\mathrm{X}-\mu) = (\mathrm{X}-\mu)^{\mathrm{T}}\left(\Psi\Psi^{\mathrm{T}}\right)^{-1}(\mathrm{X}-\mu)$$

$$= (\mathrm{X}-\mu)^{\mathrm{T}}\left(\Psi^{\mathrm{T}}\right)^{-1}(\Psi)^{-1}(\mathrm{X}-\mu) = (\mathrm{X}-\mu)^{\mathrm{T}}(\Psi^{-1})^{T}(\Psi)^{-1}(\mathrm{X}-\mu)$$

$$= \left[\left(\Psi^{-1}(\mathrm{X}-\mu)\right)\right]^{T}\left[\left(\Psi^{-1}(\mathrm{X}-\mu)\right)\right] \tag{2}$$

Where Σ is the population covariance matrix and

$$\Sigma = \Psi\Psi^{\mathrm{T}} \tag{3}$$

is the Cholesky decomposition of Σ resulting in the lower triangular matrix Ψ.

While the LHS in (2) represents distance between raw data and the data mean in statistical/input space, its equivalence in the RHS is the distance between the data point and the mean that are both transformed from the statistical space to a new vector space using Ψ^{-1} , and represented by the inner product $[\Psi^{-1}(\mathrm{X}-\mu)]^{\mathrm{T}}[\Psi^{-1}(\mathrm{X}-\mu)]$. The new vector space has following properties:

- The space has an inner product, hence it is an inner product space.
- The inner product represents the distance between the points.
- If μ = 0, the inner product represents the norm.

All these are properties of Euclidean space. Hence the new vector space where the original data was transformed using Ψ^{-1} is the Euclidean Space with the transformed data is given by

$$\mathrm{X}^{\mathrm{Euclidean}} = \Psi^{-1}\mathrm{X}^{\mathrm{Input}} \tag{4}$$

and the original statistical/input space is a non-Euclidean space.

Mahalanobis transformation uses the covariance matrix of the probability distribution, which is derived from the population/sample. In SVM of binary classification, the dataset belongs to two classes given by y=1 or y=-1. As these are two distinct labels with distinct features and parameters, they can be considered as two distinct populations, which correspond to two distinct distributions. These distributions have their unique covariance structure, hence will be characterized by their distinct covariance matrix $\Sigma_{\mathrm{y}=1}$ and $\Sigma_{\mathrm{y}=-1}$ respectively. Accordingly, the data transformation can be given by

$$\mathrm{X}_{\mathrm{y}=1}^{\mathrm{Euclidean}} = \Psi_{\mathrm{y}=1}^{-1}\mathrm{X}_{\mathrm{y}=1}^{Input} \tag{5}$$

$$\mathrm{X}_{\mathrm{y}=-1}^{\mathrm{Euclidean}} = \Psi_{\mathrm{y}=-1}^{-1}\mathrm{X}_{\mathrm{y}=-1}^{Input} \tag{6}$$

In SVM the objective is to find out a linear hyperplane that separates the data with maximum margin from the data points of either class. The equation of the hyperplane and the magnitude of the margin derived from the Euclidean distance formula and principles of cartesian coordinate system. In the Euclidean space, the equation of the maximum margin classifier becomes (considering hard margin SVM, ξ_i=0)

$$\Theta^{\mathrm{T}}\mathrm{X}^{\mathrm{Euclidean}} + \Theta_0=0 \tag{7}$$

And the margin is given by

$$\frac{1}{\sqrt{\Theta^{\mathrm{T}}\Theta}} \tag{8}$$

So the optimization problem becomes

$$\mathrm{Min}\,\frac{1}{2}\Theta^{\mathrm{T}}\,\Theta \tag{9}$$

Subject to

$$y_i(\Theta^{\mathrm{T}}\mathrm{X}^{\mathrm{Euclidean}} + \Theta_0) \geq 1 \tag{10}$$

Formulating Euclidean space-non Euclidean input space equivalence, the decision boundary given in (5) becomes (considering hard margin SVM, ξ_i=0)

$$\Theta^T \Psi_{y=1}^{-1} X_{y=1}^{\text{Input}} + \Theta_0 = 0 \qquad \text{or} \qquad \left(\left(\Psi_{y=1}^{-1} \right)^T \Theta \right)^T X_{y=1}^{\text{Input}} + \Theta_0 = 0 \tag{11}$$

for population of data points labelled y=1.

What is the margin of the linear classifier hyperplane given in (10) in the input space? If hypothetically, we apply the rules of Cartesian coordinate geometry and try to calculate the margin from the margin boundary to the classifier, the margin is

$$\frac{1}{\sqrt{\left(\left(\Psi_{y=1}^{-1} \right)^T \Theta \right)^T \left(\left(\Psi_{y=1}^{-1} \right)^T \Theta \right)}} = \frac{1}{\sqrt{\Theta^T \left(\left(\Psi_{y=1}^{-1} \right)^T \right)^T \left(\left(\Psi_{y=1}^{T} \right)^{-1} \Theta \right)}} = \frac{1}{\sqrt{\Theta^T \Psi_{y=1}^{-1} \left(\Psi_{y=1}^{T} \right)^{-1} \Theta}} = \frac{1}{\sqrt{\Theta^T \left(\Psi_{y=1} \right)^{-1} \left(\Psi_{y=1}^{T} \right)^{-1} \Theta}} =$$
$$\frac{1}{\sqrt{\Theta^T \left(\Psi_{y=1}^{T} \Psi_{y=1} \right)^{-1} \Theta}} = \frac{1}{\sqrt{\Theta^T \left(\Psi_{y=1}^{-1} \Psi_{y=1} \Psi_{y=1}^{T} \Psi_{y=1} \right)^{-1} \Theta}} = \frac{1}{\sqrt{\Theta^T \left(\Psi_{y=1}^{-1} \Sigma_{y=1} \Psi_{y=1} \right)^{-1} \Theta}} =$$
$$\frac{1}{\sqrt{\Theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \Theta}} \tag{12}$$

Hence the margin maximization problem in the input space for y=1 becomes

$$\text{Maximize } \frac{1}{\Theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \Theta} \tag{13}$$

Satisfying the constraint

$$\Theta^T \Psi_{y=1}^{-1} X_{y=1}^{\text{Input}} + \Theta_0 \geq 1 \tag{14}$$

Similarly for the data label y=-1, the margin maximization problem becomes

$$\text{Maximize } \frac{1}{\Theta^T \Psi_{y=-1}^{-1} \Sigma_{y=-1}^{-1} \Psi_{y=-1} \Theta} \tag{15}$$

Satisfying the constraint

$$\Theta^T \Psi_{y=-1}^{-1} X_{y=-1}^{\text{Input}} + \Theta_0 \leq -1 \tag{16}$$

Equations (17), (18), (19), (20) can be combined to form a unified optimization problem where the objective function is

$$\text{Maximize } z^* \frac{1}{\Theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \Theta} + (1-z)^* \frac{1}{\Theta^T \Psi_{y=-1}^{-1} \Sigma_{y=-1}^{-1} \Psi_{y=-1} \Theta}$$

Satisfying the constraint

$$z^*(\Theta^T \Psi_{y=1}^{-1} X_{y=1}^{\text{Input}} + \Theta_0 \geq 1) + (1-z)^*(\Theta^T \Psi_{y=-1}^{-1} X_{y=-1}^{\text{Input}} + \Theta_0 \leq -1)$$

where $\quad z = 1 \quad \forall$ y=1

$\qquad 0 \quad$ Otherwise

Comparison of maximized margins obtained from equations (14) between classifiers of y=1 and y=-1 result in

$$\frac{\mathrm{Margin}_{y=1}}{\mathrm{Margin}_{y=-1}} = \sqrt{\frac{\theta^T \Psi_{y=-1}^{-1} \Sigma_{y=-1}^{-1} \Psi_{y=-1} \theta}{\theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \theta}} \quad = f\left(\frac{\theta^T \Psi_{y=-1}^{-1} \Sigma_{y=-1}^{-1} \Psi_{y=-1} \theta}{\theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \theta}\right) \quad = \quad f\left(\frac{\Sigma_{y=-1}^{-1}}{\Sigma_{y=1}^{-1}}\right) \qquad (17)$$

Since these operations are valid in the Euclidean space only, it shows that

"Principles of support vector classification (calculation of the maximum margin and identification of maximum margin hyperplane) are optimal only when the data is transformed from the input/statistical space to the Euclidean space using Ψ^{-1}, which is inverse of the lower triangular matrix obtained by Cholesky Decomposition of the population covariance matrix. The SVM optimization problem, loss function and constraints are formulated in the transformed space."

As we are maximizing $\frac{1}{\theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \theta}$, not $\frac{1}{\sqrt{\theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \theta}}$ in the optimization problem, and the presence of slack variables ensures parameter values while maximizing $\frac{1}{\theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \theta}$ are different from maximizing $\frac{1}{\sqrt{\theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \theta}}$. Since we are using parameter values obtained during maximizing $\frac{1}{\theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \theta}$ during soft margin classification, then the presence of slack variable means

$$\frac{\mathrm{Margin}_{y=1}}{\mathrm{Margin}_{y=-1}} = \frac{\theta^T \Psi_{y=-1}^{-1} \Sigma_{y=-1}^{-1} \Psi_{y=-1} \theta}{\theta^T \Psi_{y=1}^{-1} \Sigma_{y=1}^{-1} \Psi_{y=1} \theta}.$$

There are two sets of optimization problems to be solved. This gives rise to the following hypotheses:

Hypothesis 3.1. For a two-class problem, the application of SVM in the input space domain generates not one, but two unique optimization problem formulations resulting in two unique linear classifiers—each input space having its own linear classifier. In an N-class problem, there will be N data class distributions and N input spaces; hence, there will be N linear classifiers.

Hypothesis 3.2. In the input space, margin for each data class is a function of respective population covariance matrix, $\frac{1}{\Sigma^{-1}}$. Hence the assumptions of KKT boundary conditions are not optimal in the input space as each data point (other than support vectors) contributes to Σ.

Hence unlike the original support vector machine algorithm, in a Non-Euclidean space the margin of the classifier is dependent on the intra-class data covariance (variance is essentially covariance of a random variable with itself).

Comparison with other works

In addition, various studies have been carried out to incorporate data covariance via Mahalanobis distance in SVM optimization problem. The works of Tsang et al [1] Ke et al [2] Peng and Xu [3] have led to the following set of optimization equations:

$$\text{Min } \frac{1}{2}\theta^T \Sigma^{-1} \Theta + \lambda \sum_{i=1}^{N} \xi_i \qquad (1)$$

$$\text{s.t. } y_i(\theta^T \Sigma^{-1} x_i + \theta_0) \geq 1\text{-}\xi_i, \qquad \xi_i \geq 0 \qquad \forall i = 1,2,\dots,N \qquad (2)$$

while Huang et.al. [4] and Zafeiriou et al. [5] arrived at another set of equations:

$$\text{Min } \frac{1}{2}\theta^T\Sigma^{-1}\Theta + \lambda\sum_{i=1}^{N}\xi_i \quad (3)$$

$$\text{s.t. } y_i(\theta^T x_i + \theta_0) \geq 1\text{-}\xi_i,\ \xi_i \geq 0 \quad \forall i = 1,2,....N \quad (4)$$

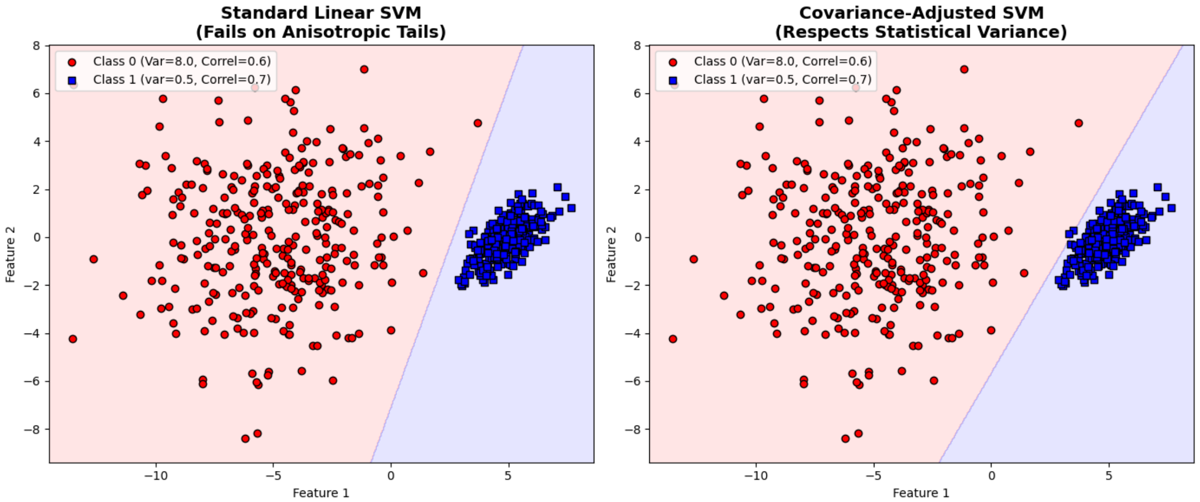
with Zafeiriou et. al using scatter matrix instead of covariance matrix. While equations (1) to (4) have tried to incorporate the data covariance into optimization problem of SVMs with significant improvement in performance w.r.t. linear SVMs, they suffer from the following limitations:

a) It is not clear how they chose equations (1), (2) or (3). The derivation of variance adjusted objective function and constraints based on first principles are not clear in the studies;
b) There is vector space inconsistency in equations (3) and (4) i.e. equation (3) is in Euclidean space while (4) is in non-Euclidean space.
c) There is dimensional inconsistency in the constraint equation (2). As per Ke et al. [6], Mahalanobis distance is Euclidean distance after decorrelation and standardization of every random variable. If we consider the special case where correlation between variables is 0, then each variable is being standardized by the formula $\frac{X}{\sigma^2}$. However, in univariate setting, standardization of random variables is carried out by using the formula $\frac{(X)}{\sigma}$. Hence in multivariate setting the standardization the random variables need to be carried out using $\Sigma^{\frac{-1}{2}}$, not Σ^{-1}. This is leading to dimensional inconsistency.

In this study, by starting from first principles and focusing on Mahalanobis Distance as a vector space transformation, we formulate the optimization problem in CSVM keeping it both vector space and dimensionally consistent, and thus address the limitations of previous studies.

Simulated Study to show how CSVM decision boundary considers variance and covariance effects

For this simulation we created two synthetic distributions- a Gaussian with mean=5.0 and covariance matrix = $\begin{bmatrix} 0.5 & 0.7 \\ 0.7 & 0.5 \end{bmatrix}$ and another Gaussian distribution with mean=-5.0 and covariance matrix = $\begin{bmatrix} 8.0 & 0.6 \\ 0.6 & 8.0 \end{bmatrix}$ and fitted linear SVM and CSVM. It can be seen that CSVM decision boundary is closer to the distribution with lower variance. In addition, the decision boundary is also taking care of the data correlations.



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